

# CE 310

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## Week 9 — Simple Linear Regression

From Correlation to Prediction

University of Arizona · Dr. Wooyoung Jung · Fall 2026

# Today's Roadmap

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By the end of this lecture you will be able to:

1. Explain why correlation alone is insufficient for engineering decisions
2. Write and interpret the regression equation  $\hat{y} = \beta_0 + \beta_1 x$
3. Interpret slope and intercept in physical units
4. Read  $R^2$  as the fraction of variance explained
5. Use a residual plot to identify where a linear model breaks down
6. Connect linear regression to supervised machine learning

**Dataset:** DOE Reference Medium Office, Tucson — 8,760 hourly observations (same as Weeks 5–7)

# Python Readiness — What You Need This Week

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## Assumed from Weeks 5–8:

- `pandas` — read CSV, column access, filtering
- `matplotlib.pyplot` — scatter plots, line plots, labels
- `numpy` — array arithmetic, `mean()`, `std()`

## New this week:

- `scipy.stats.linregress` — fits OLS line; returns slope, intercept, r, p-value, SE
- `statsmodels.formula.api` — prediction intervals (tutorial only)

If Weeks 5–8 labs feel shaky, revisit them before the tutorial and assignment.

# Week 7 Bridge — What Correlation Left Unanswered

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In Week 7 you computed:  $r(\text{Outdoor\_Temp\_F}, \text{Cooling\_kWh}) = 0.78$

## What $r$ tells you:

- Strong positive linear relationship ✓
- Outdoor temperature and cooling move together ✓
- Dimensionless — works across any units ✓

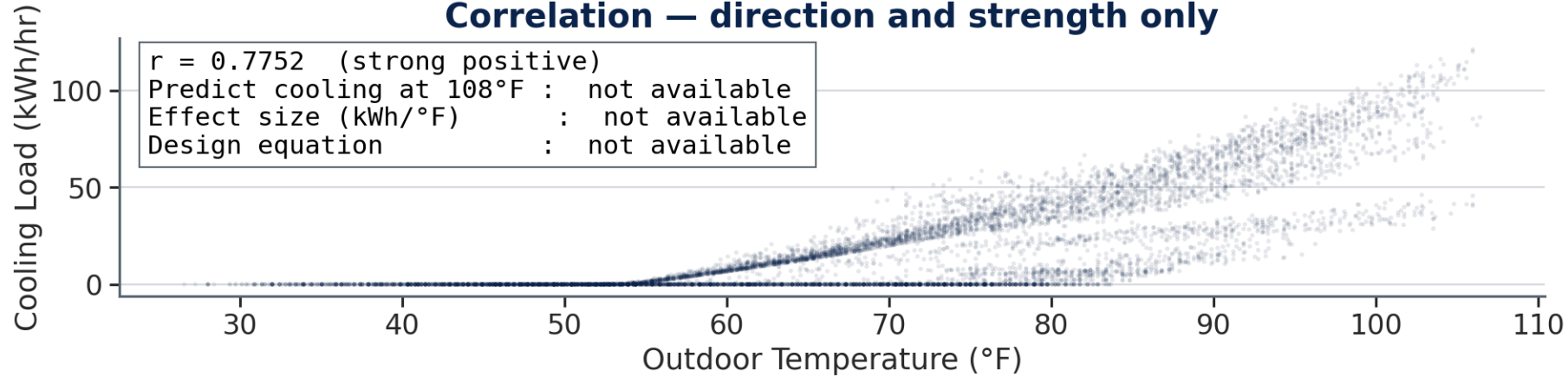
## What $r$ cannot tell you:

- How many kWh per 1°F rise?
- What is cooling demand at 108°F design temperature?
- At what outdoor temperature is the chiller off?

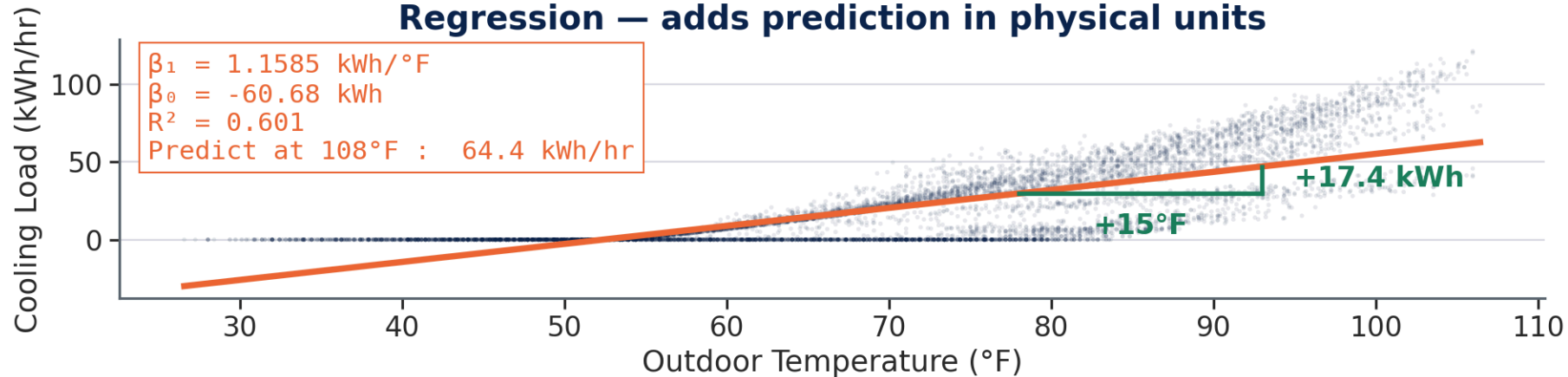
These questions require a **regression equation**.

# From Correlation to Prediction

## Correlation — direction and strength only



## Regression — adds prediction in physical units



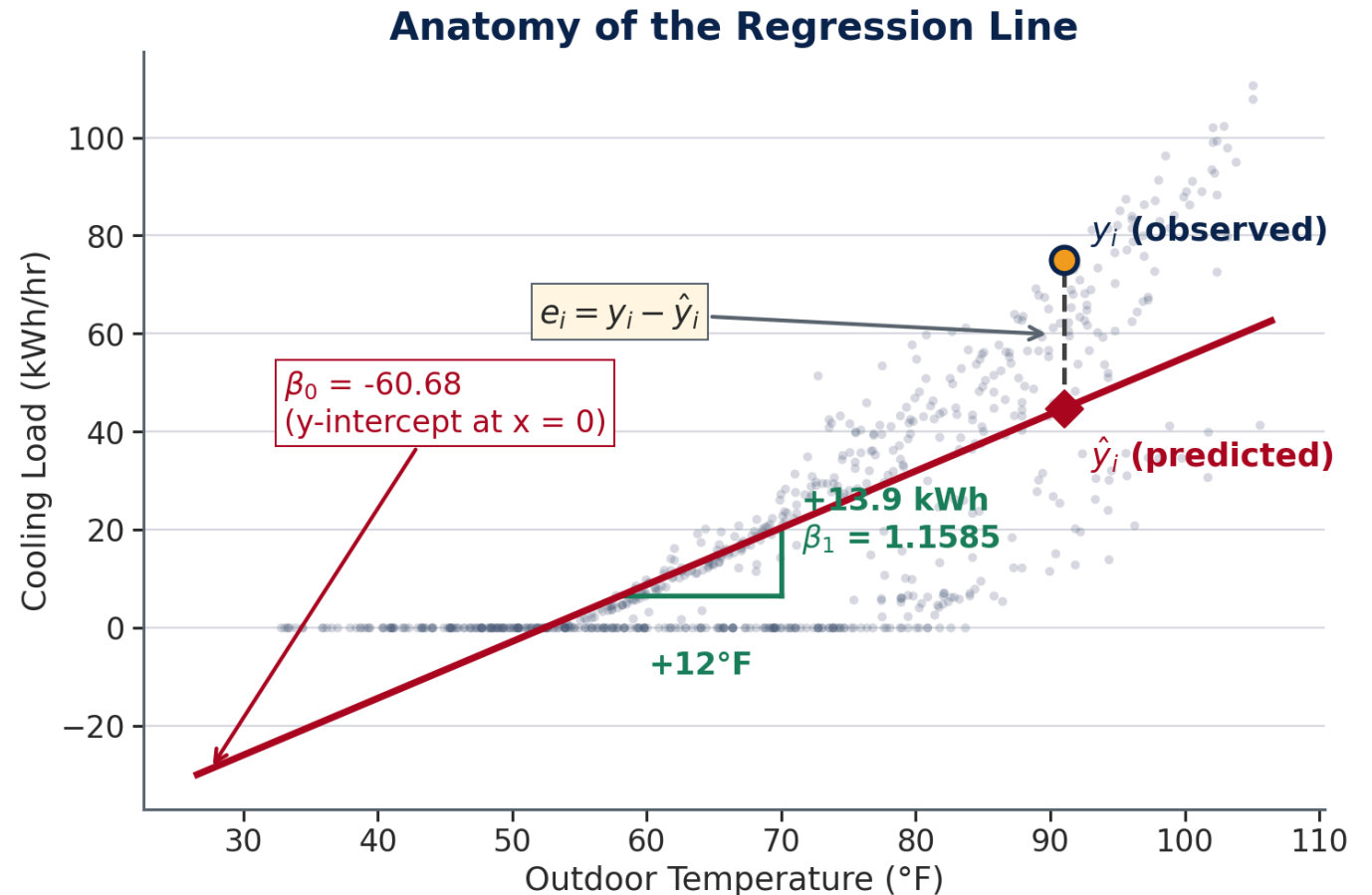
# The Regression Equation

$$\hat{y} = \beta_0 + \beta_1 \cdot x$$

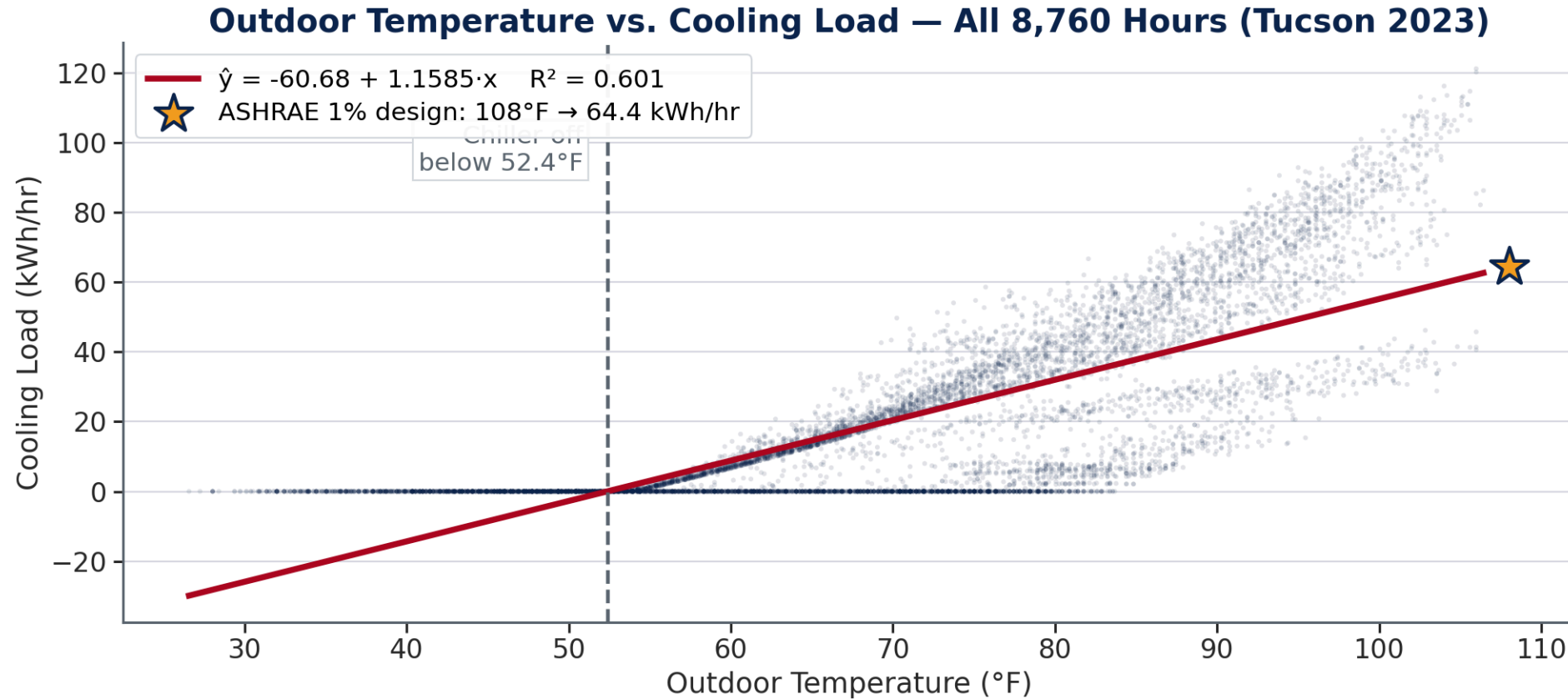
- $\hat{y}$  — predicted value (not the observed value)
- $\beta_1$  — slope: change in  $\hat{y}$  per 1-unit rise in  $x$
- $\beta_0$  — intercept: predicted  $\hat{y}$  when  $x = 0$
- $x$  — predictor (input) variable

**Tucson cooling model:**

$$\widehat{\text{Cooling}} = -60.68 + 1.1585 \times T_{\text{out}}$$



# The Fitted Model — 8,760 Hours



$\beta_1 = 1.1585 \text{ kWh/°F} \cdot R^2 = 0.601 \cdot n = 8,760 \text{ hrs} \cdot \text{DOE Reference Medium Office, Tucson AZ 2023}$

# Interpreting the Slope

$$\beta_1 = 1.1585 \text{ kWh/}^\circ\text{F}$$

"For every 1°F rise in outdoor temperature, predicted cooling demand increases by **1.16 kWh/hr.**"

## Engineering applications of the slope:

| Scenario                       | Calculation        | Result                           |
|--------------------------------|--------------------|----------------------------------|
| 10°F hotter design day         | $1.1585 \times 10$ | +11.6 kWh/hr more cooling        |
| Summer vs. mild day (+20°F)    | $1.1585 \times 20$ | +23.2 kWh/hr more cooling        |
| Climate zone shift (+5°F mean) | $1.1585 \times 5$  | +5.8 kWh/hr annual load increase |

The slope is the number a mechanical engineer needs to size cooling equipment for different climate conditions.



## $r$ vs. $\beta_1$ — Two Different Questions

|                   | Pearson correlation $r$   | Regression slope $\beta_1$       |
|-------------------|---------------------------|----------------------------------|
| Measures          | Strength and direction    | Rate of change                   |
| Units             | Unitless ( $-1$ to $+1$ ) | kWh / °F (physical units)        |
| Question answered | Is there a relationship?  | How much does $y$ change per °F? |
| Tucson example    | $r = 0.78$                | $\beta_1 = 1.16$ kWh / °F        |

A strong  $r$  does not imply a steep  $\beta_1$  — and vice versa.

$r$  tells you the relationship is real and strong.

$\beta_1$  tells you what to do with it — write a design equation, size equipment, predict demand.

# Interpreting the Intercept

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$$\beta_0 = -60.68 \text{ kWh}$$

The intercept is "predicted cooling when outdoor temperature = 0°F."

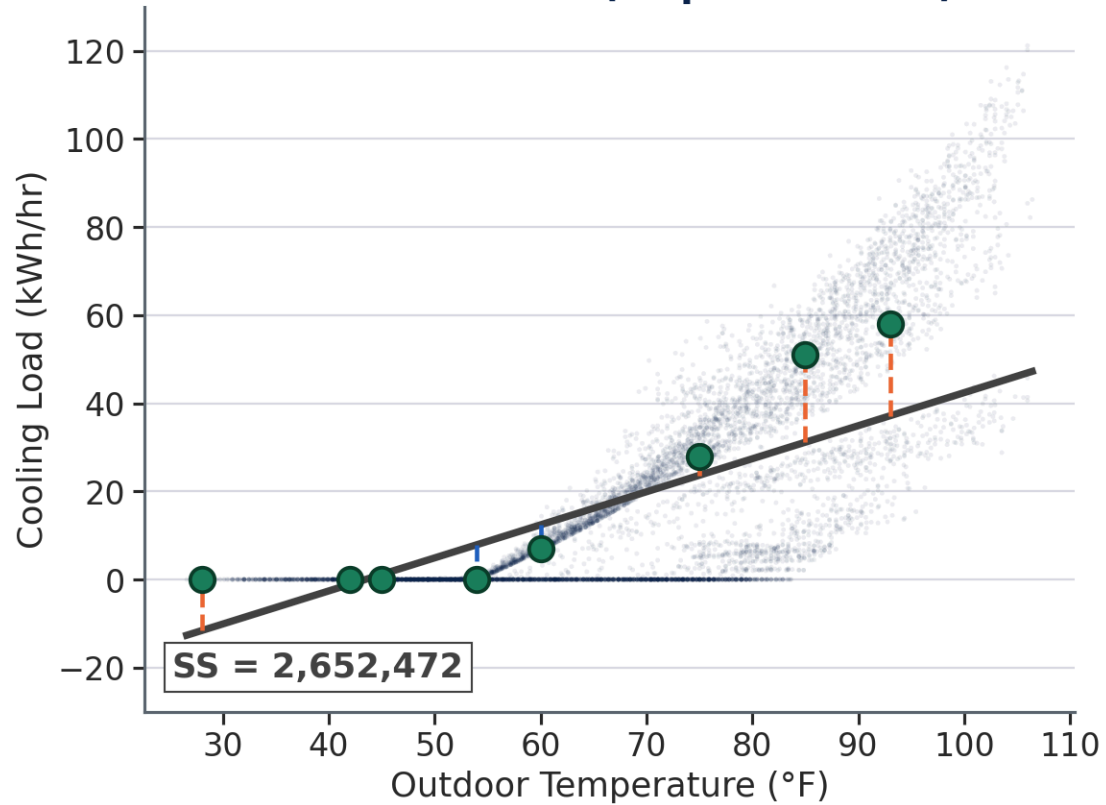
- Tucson's minimum observed temperature: **26.6°F** — never reaches 0°F
- The intercept is mathematically valid but **physically meaningless** for this dataset
- A negative intercept simply reflects that the chiller is off in winter

⚠ Never interpret an intercept at  $x = 0$  if 0 is outside your observed data range. Extrapolation is unreliable and often physically absurd.

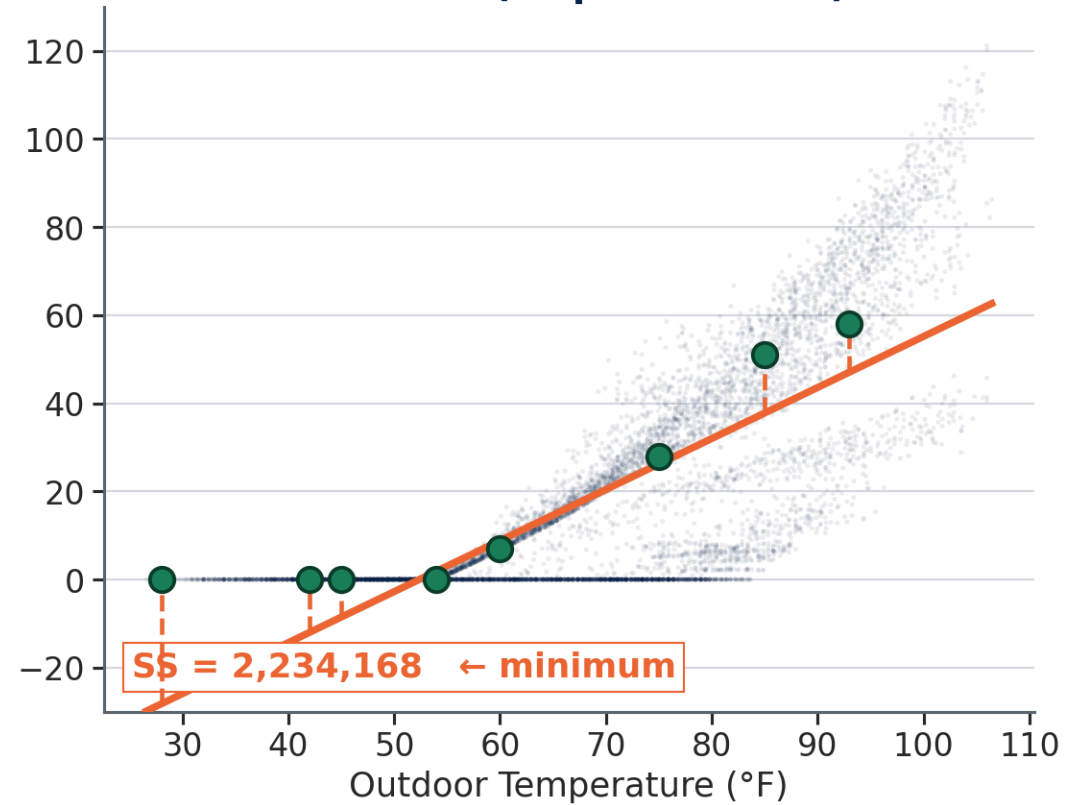
# OLS Minimizes Total Squared Error

OLS minimizes  $\sum e_i^2$  — the sum of squared vertical distances ( $n = 8,760$ )

Candidate line (slope = 0.7500)



OLS line (slope = 1.1585)



# How the Line Is Found — Least Squares

Choose  $\beta_0$  and  $\beta_1$  to **minimize total squared error** — where  $e_i = y_i - \hat{y}_i$ :

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

**Why squaring?** A 10 kWh miss counts 100× more than a 1 kWh miss; squaring also yields a unique closed-form solution (minimizing  $|e_i|$  does not).

**Week 7 connection — closed-form solution:**

$$\hat{\beta}_1 = r \cdot \frac{\sigma_Y}{\sigma_X} = \frac{\text{cov}(X, Y)}{\text{var}(X)}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

The line always passes through  $(\bar{x}, \bar{y})$ . Verify:  $-60.68 + 1.1585 \times 68.7 = 19.0$  kWh/hr ✓

## Quick Gut Check — $r$ to $R^2$

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From Week 7:  $r = 0.78$  between outdoor temperature and cooling demand.

**What fraction of cooling variance does outdoor temperature explain?**

**A)**  $\approx 20\%$       **B)**  $\approx 40\%$       **C)**  $\approx 60\%$       **D)**  $\approx 80\%$

*Work it out — you have the number you need from Week 7.*

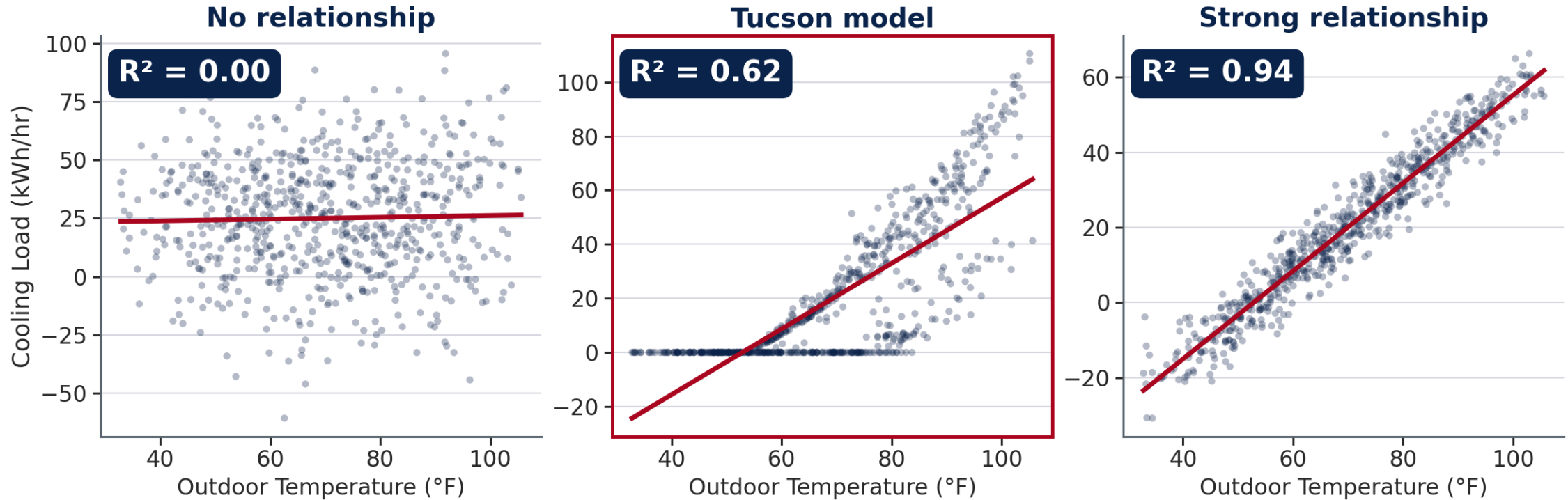
✓  $R^2 = r^2 = 0.78^2 \approx 0.60$

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Outdoor temperature explains **60%** of cooling variance.

- The remaining **40%** — occupancy patterns, solar gain, equipment loads — is not captured by temperature alone
- $R^2 = r^2$  holds **only for simple linear regression** (one predictor); in Week 10 with two predictors it no longer applies
- Adding business-hours as a second predictor next week pushes  $R^2$  to **0.91** during occupied hours

# $R^2$ — Fraction of Variance Explained

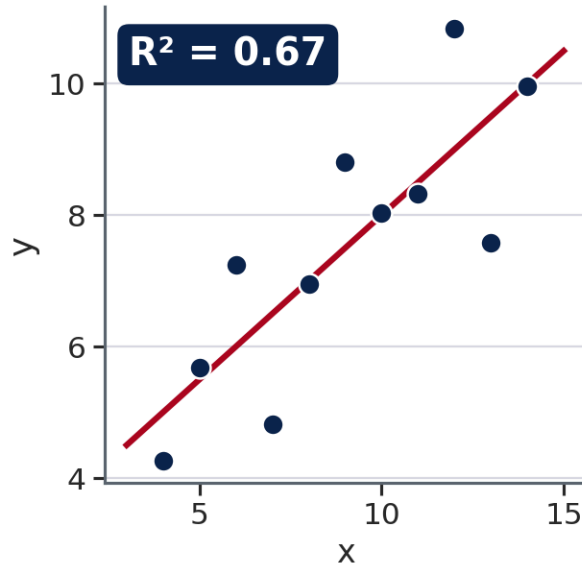


**Tucson:  $R^2 = 0.601$**  — outdoor temperature explains 60% of cooling variance; the rest is occupancy, solar, and equipment.

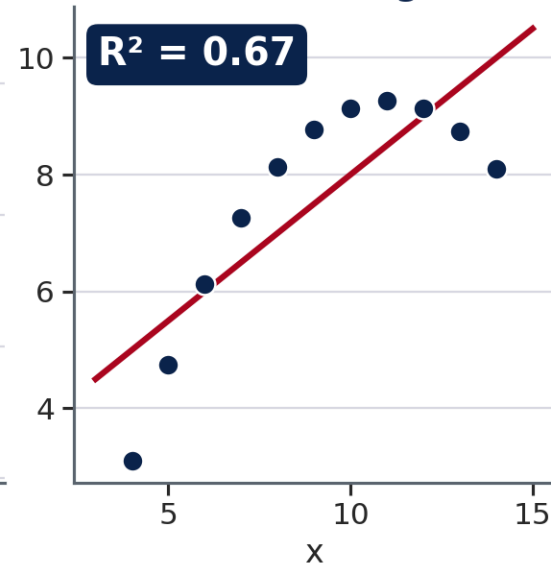
# Anscombe's Quartet — Same Statistics, Different Relationships

Anscombe's Quartet — nearly identical statistics, completely different relationships

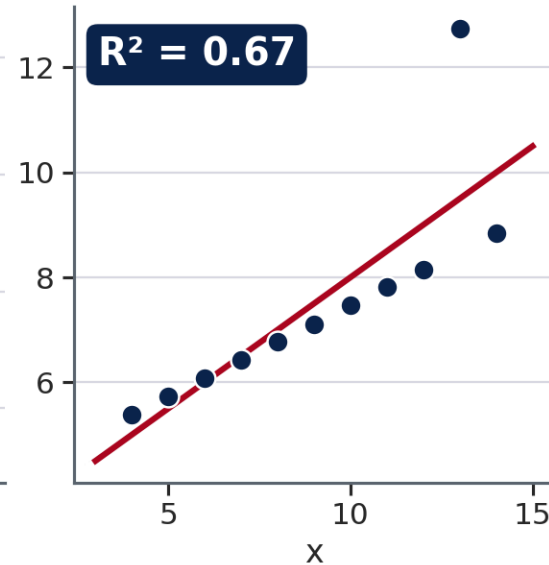
Dataset I:  
Ideal — linear



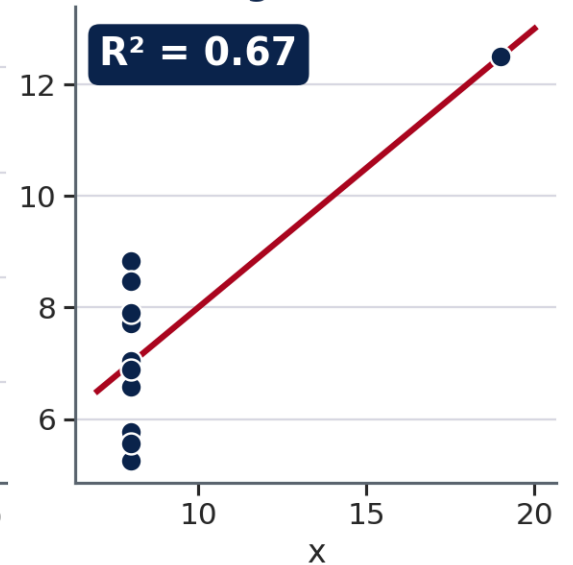
Dataset II:  
Curved — wrong form



Dataset III:  
Outlier — distorts line



Dataset IV:  
Leverage — no trend





# What Anscombe's Quartet Tells Us

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All four datasets share nearly identical statistics:

**slope  $\approx 0.50$ , intercept  $\approx 3.0$ ,  $R^2 \approx 0.67$**

Summary statistics alone never replace a scatter plot. Always visualize your data before fitting any model.

$R^2$  cannot tell you that your model is wrong. Only the scatter plot and residual plot can.

- **Dataset I** — the only one where OLS is appropriate
- **Dataset II** — curved relationship; OLS underpredicts in the middle, overpredicts at the extremes
- **Dataset III** — one outlier inflates the slope; remove it and  $R^2 > 0.99$
- **Dataset IV** — no real relationship; a single leverage point creates the illusion of a trend

# Regression Assumptions — LINE

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For OLS estimates to be valid and confidence intervals to be trustworthy, four conditions must hold:

- **L — Linearity**: the true relationship between  $x$  and  $y$  is linear
- **I — Independence**: each observation is independent (no serial correlation in residuals)
- **N — Normality**: residuals follow an approximately normal distribution
- **E — Equal variance**: residual spread is constant across all values of  $x$  (homoscedasticity)

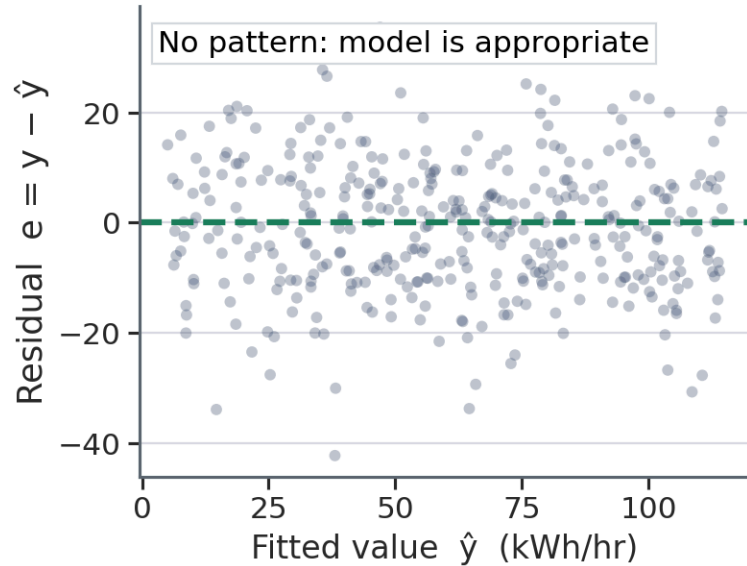
The residual patterns figure (next slide) shows what **L** and **E** violations look like in practice.

The Tucson hourly data has mild **I** violations (consecutive hours, same building) — a limitation accepted at this course level.

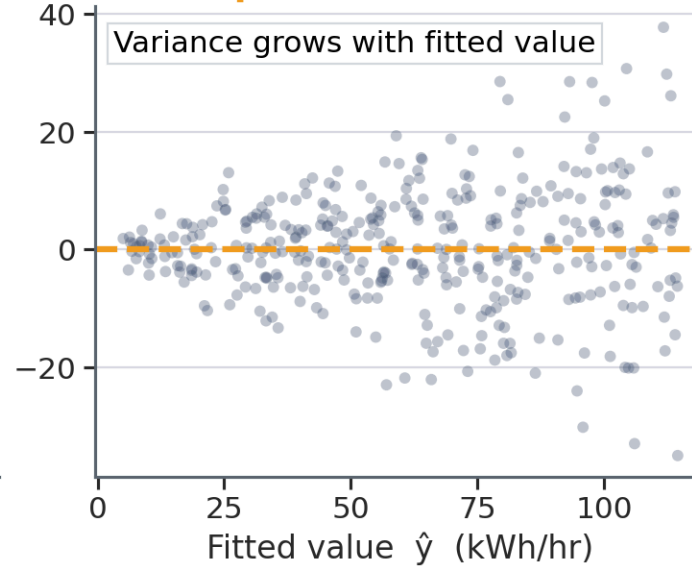
# Residual Patterns — What to Look For

Residual plot patterns — what each shape tells you about the model

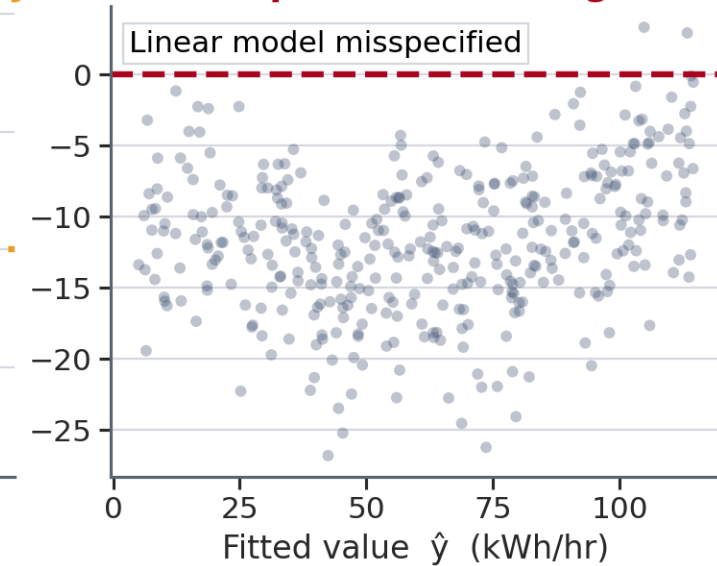
Good — random scatter



Fan shape — heteroscedasticity



Curved pattern — wrong form



# Residuals — Checking the Model

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**Residual:**  $e_i = y_i - \hat{y}_i$

- **Positive residual:** actual > predicted (model underestimates)
- **Negative residual:** actual < predicted (model overestimates)

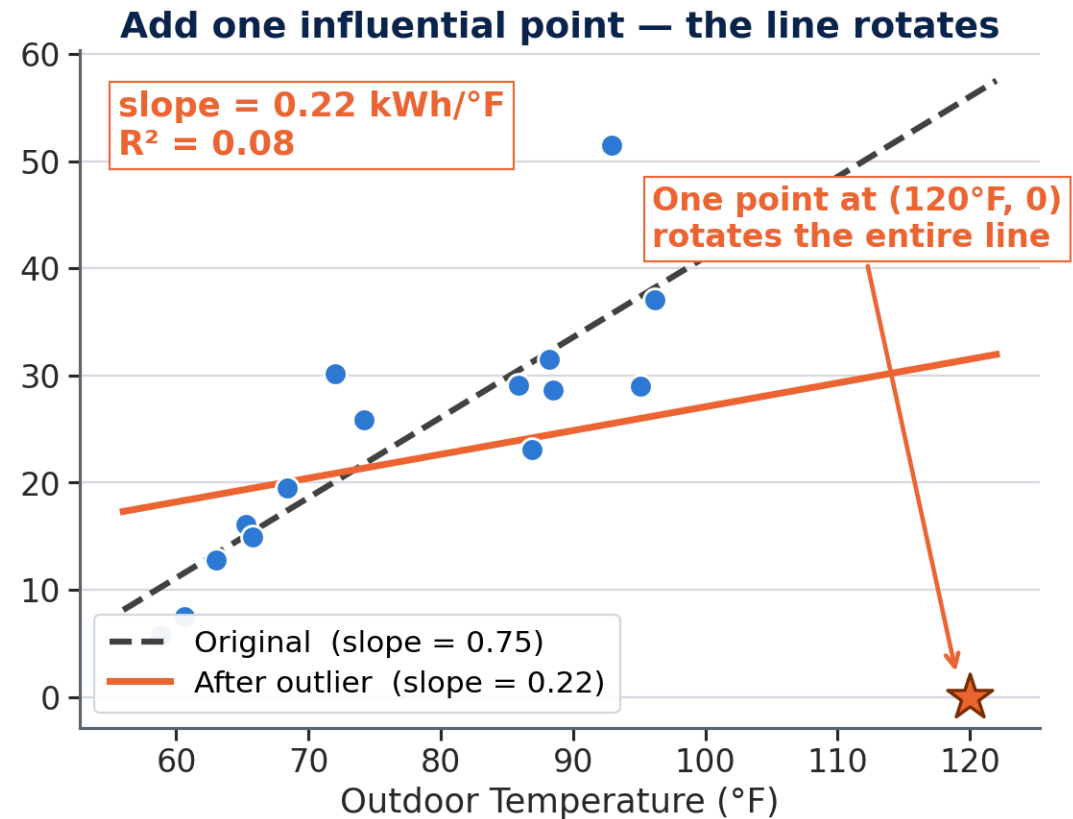
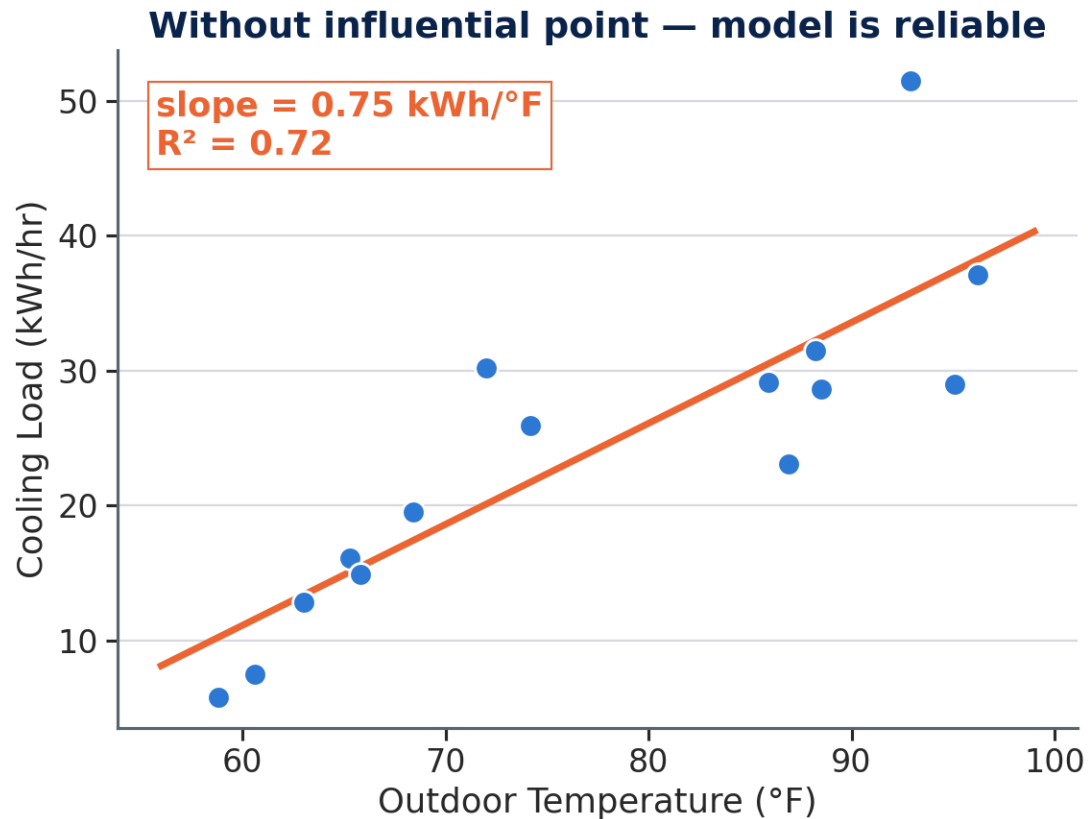
**What a good residual plot looks like:**

- Residuals scattered randomly above and below zero
- No obvious pattern across the range of x

**What signals a problem:**

- Systematic positive or negative residuals in a particular x-range → model is biased in that region
- Fan-shaped spread → the model becomes less reliable at high temperatures

# Outliers and Influential Points



One observation at (120°F, 0 kWh) cut the slope from **0.78 to 0.25 kWh/°F** — making the model nearly useless.

# Outliers – Three Concepts

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| Term                       | What it means                           | Effect on OLS                         |
|----------------------------|-----------------------------------------|---------------------------------------|
| <b>Outlier</b>             | Unusual $y$ value at a typical $x$      | Inflates residual sum; may bias slope |
| <b>High-leverage point</b> | Extreme $x$ value (far from $\bar{x}$ ) | Pulls the line toward itself          |
| <b>Influential point</b>   | Unusual $y$ AND extreme $x$             | Can completely change the slope       |

An influential point combines both conditions — the figure above is an example.

Always inspect your data for extreme points before reporting any model. In small datasets, one bad sensor reading or data entry error can invalidate the entire analysis.

# Tucson Cooling Model

Putting the numbers together

# The Fitted Model

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$$\overline{\text{Cooling\_kWh}} = -60.68 + 1.1585 \times \text{Outdoor\_Temp\_F}$$

## Model statistics:

- Slope  $\beta_1 = \mathbf{1.1585 \text{ kWh/}^\circ\text{F}}$
- Intercept  $\beta_0 = \mathbf{-60.68 \text{ kWh}}$
- $R^2 = \mathbf{0.601}$
- p-value  $\approx 0$  — if temperature had no real effect, a slope this large would be essentially impossible by chance (Week 11 formalizes this)
- $n = 8,760$  hourly observations

## Sanity check — plug in the mean:

- Mean outdoor temp:  $\bar{x} = 68.7^\circ\text{F}$
- Predicted:  $-60.68 + 1.1585 \times 68.7 = \mathbf{19.0 \text{ kWh/hr}}$
- Actual mean cooling:  $\mathbf{19.0 \text{ kWh/hr } \checkmark}$

The regression line always passes through  $(\bar{x}, \bar{y})$ .



# Prediction at Design Conditions

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## CE Track — ASHRAE 1% Cooling Design

- Design temperature: **108°F** (top 1% of Tucson hours)
- Prediction:
  - $-60.68 + 1.1585 \times 108 = \mathbf{64.4 \text{ kWh/hr}}$
- Annual mean: 19.0 kWh/hr
- **Peak-to-mean ratio: 3.4×**

Sizing for the mean would leave the building severely under-cooled on hot summer days.

## ArcE Track — Economizer Cutoff

- ASHRAE 90.1 economizer limit: **75°F** (Climate Zone 2B)
- Below 75°F: outside air provides free cooling
- Prediction:
  - $-60.68 + 1.1585 \times 75 = \mathbf{26.2 \text{ kWh/hr}}$
- Above 75°F: mechanical cooling required every hour

# Regression Appears Across CE Disciplines

The same model form recurs wherever one continuous quantity drives another.

| Discipline              | Predictor $x$            | Response $y$             | Slope meaning                  |
|-------------------------|--------------------------|--------------------------|--------------------------------|
| <b>HVAC</b> (this week) | Outdoor temp (°F)        | Cooling load (kWh/hr)    | +1.16 kWh per °F               |
| <b>Structural</b>       | Applied load (kN)        | Mid-span deflection (mm) | spring stiffness — Hooke's Law |
| <b>Transportation</b>   | Daily traffic (vehicles) | Pavement wear (ESAL/yr)  | load damage factor             |
| <b>Water resources</b>  | Annual rainfall (in)     | Annual runoff (in)       | runoff coefficient             |

The regression you learned this week is not an HVAC tool — it is a universal CE tool. The method travels; only the domain vocabulary changes.

# Uncertainty in the Prediction

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The slope 1.1585 kWh/°F is an **estimate** — computed from 8,760 observations with noise. Every estimate has uncertainty. `scipy.stats.linregress` returns the **standard error of the slope**.

One call fits the line and returns the slope, intercept,  $r$ ,  $p$ -value and the **standard error of the slope** — all five at once.

| Quantity                    | Value                        |
|-----------------------------|------------------------------|
| Slope $\beta_1$             | 1.1585 kWh/°F                |
| Standard error of $\beta_1$ | 0.0101                       |
| 95% confidence interval     | <b>1.139 to 1.178 kWh/°F</b> |

The interval is  $\beta_1 \pm 1.96 \times \text{SE}$ . It is narrow because  $n = 8,760$  — with that much data the slope is pinned down tightly even though  $R^2$  is only 0.601.

*The tutorial covers how to call it and read the output.*

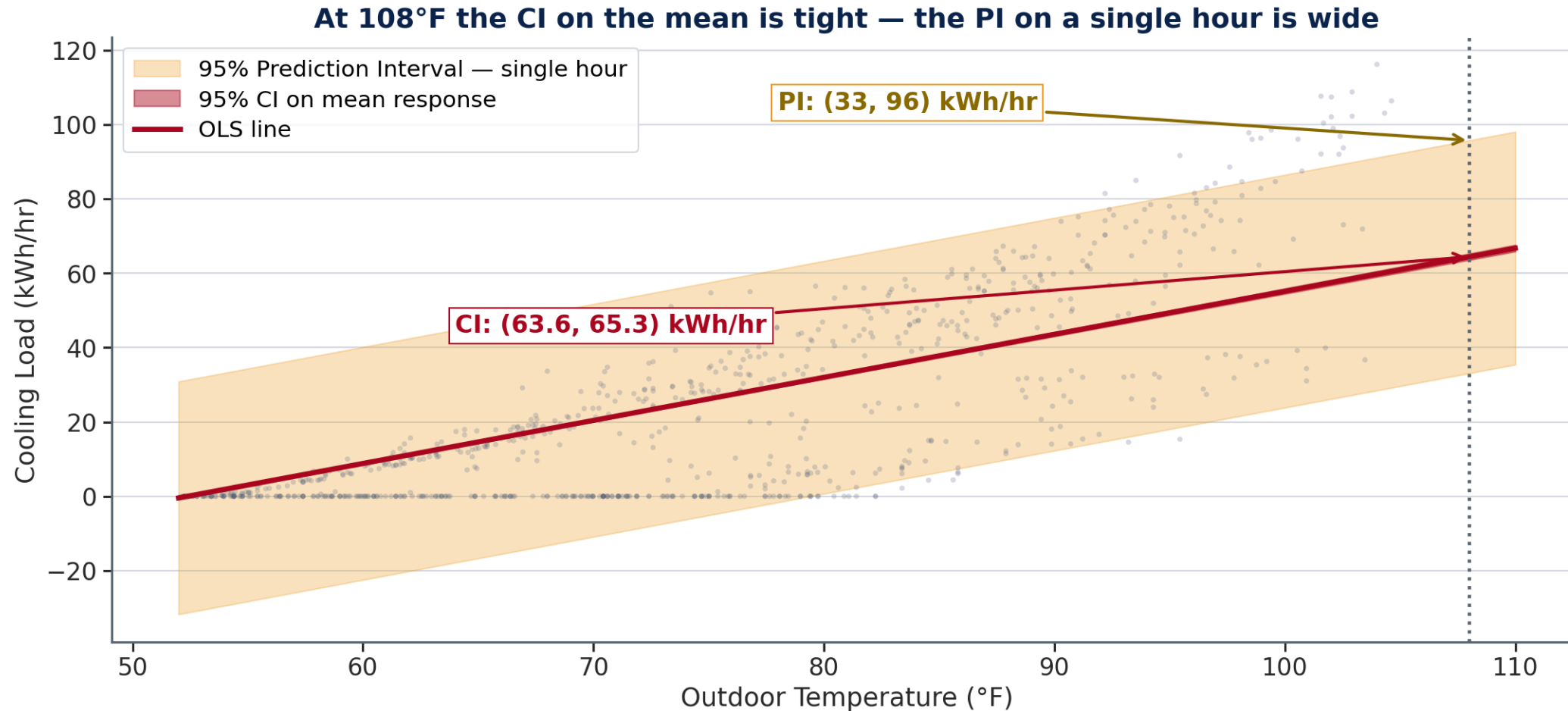
## Uncertainty in the Prediction — Interpretation

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**Interpretation:** We are 95% confident the true slope lies between **1.139 and 1.178 kWh/°F**.  
At 108°F, this 0.039 range translates to only  $\pm 0.8$  kWh/hr of slope uncertainty in the final prediction.

💡 The slope CI is narrow because  $n = 8,760$  is large. But the uncertainty in a *single predicted hour* is far larger — see the next slide.

# Confidence Band vs. Prediction Band



# Prediction Interval — The Engineering Deliverable

Two bands exist around a regression line. For equipment sizing, only one matters.

| Band                    | What it bounds                                              | At 108°F                           |
|-------------------------|-------------------------------------------------------------|------------------------------------|
| 95% CI on mean          | Where the <b>average</b> cooling falls for many 108°F hours | <b>(63.6, 65.2) kWh/hr</b> — tight |
| 95% Prediction Interval | Where a <b>single future 108°F hour</b> is likely to fall   | <b>(30, 99) kWh/hr</b> — wide      |

**Why so wide?**  $R^2 = 0.601$  means 40% of variance is unexplained — occupancy, solar gain, plug loads, equipment schedules. Any single hot hour could produce 30 to 99 kWh of cooling demand.

⚠ A chiller sized at 64 kWh/hr matches the point prediction but fails the upper end of the prediction interval. Sizing to the *mean response* means the system is undersized half the time it hits 108°F. The prediction interval upper bound — not the point estimate — is the engineering conservative case.

# Design Decision — Which PI Bound?

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90 seconds, then discuss

The 95% **Prediction Interval** at 108°F is (33, 96) kWh/hr.

You are sizing a new chiller. The unit costs **\$800 per kWh of capacity**.

**Question A:** Which bound do you use to size the chiller? Why?

**Question B:** What is the dollar cost of choosing the upper bound vs. the lower bound?

## ✓ Use the Upper PI Bound — 96 kWh/hr

**A — Size to the upper bound: 96 kWh/hr** (conservative design)

| Design choice                     | Capacity         | Capital cost    | Risk                                      |
|-----------------------------------|------------------|-----------------|-------------------------------------------|
| Lower bound (aggressive)          | 33 kWh/hr        | \$26,400        | Near-certain failure at peak              |
| Mean estimate                     | 64 kWh/hr        | \$51,200        | Fails ~50% of peak hours                  |
| <b>Upper bound (conservative)</b> | <b>96 kWh/hr</b> | <b>\$76,800</b> | Low — designed for 97.5th-percentile hour |

**B — Cost of conservatism:**  $\$76,800 - \$26,400 = \$50,400$

That \$50,400 is the insurance premium against a heatwave failure in July.

In professional practice: engineers size to the upper bound, then apply an additional safety factor on top.



# Where the Model Breaks Down

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**Physical constraint:** cooling energy cannot go below 0 kWh (chiller off = no cooling).

**The linear model predicts negative values** when:

$$-60.68 + 1.1585 \times T < 0 \implies T < 52.4^\circ\text{F}$$

**In the dataset:**

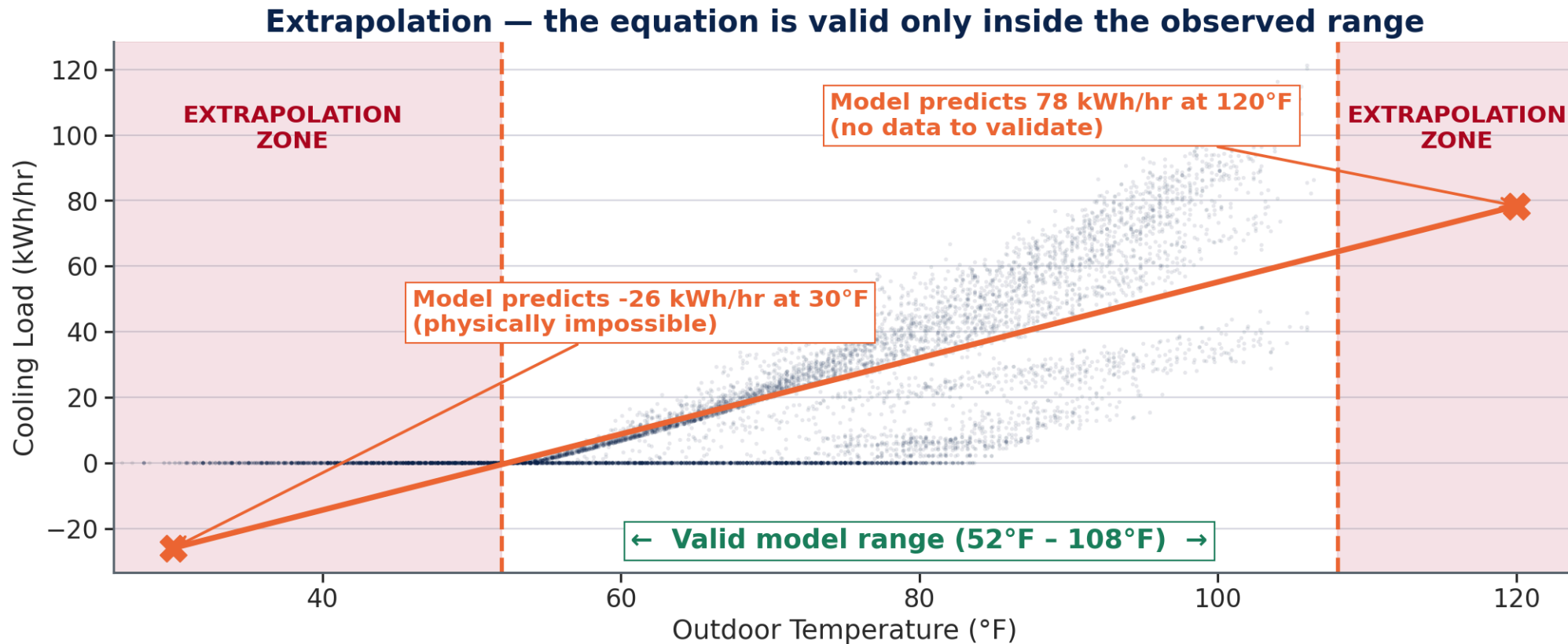
- 3,702 hours with Cooling\_kWh = 0 (chiller off, ~42% of year)
- All occur when outdoor temp < ~52°F
- Model predicts negative cooling for those hours

**What the residual plot shows:**

- At  $T < 45^\circ\text{F}$ : all residuals are **positive**
- Actual = 0, Model predicts negative  $\rightarrow$  residual = 0 – (negative) > 0
- Systematic positive bias at low temperatures

This is a structural failure of the linear model, not noise.

# Extrapolation — When the Line Lies



Use the model only within the **observed data range** ( $\approx 52^{\circ}\text{F} - 108^{\circ}\text{F}$ ). Outside that range, the equation has no physical foundation.

# Improving the Model — Preview of Week 10

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**One fix:** condition on occupancy before fitting.

| Model                | R <sup>2</sup> | What it captures                                 |
|----------------------|----------------|--------------------------------------------------|
| All hours (all-year) | 0.601          | Temperature effect, mixed with chiller-off hours |
| Business hours only  | <b>0.910</b>   | Temperature effect with occupancy held constant  |
| Summer only          | 0.562          | Temperature effect, narrower range               |

During business hours, internal heat sources — lights, computers, people — and occupancy are roughly constant — outdoor temperature explains **91%** of the residual variance.

**Week 10:** add `BusinessHours` as a second predictor explicitly:

$$\hat{y} = \beta_0 + \beta_1 \cdot \text{Outdoor\_Temp\_F} + \beta_2 \cdot \text{BusinessHours}$$

# From Regression to Machine Learning

The same idea at larger scale

# The AI Landscape — Where Regression Fits

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**Artificial Intelligence** — techniques that let computers emulate human behavior: learn, decide, recognize patterns, solve problems.

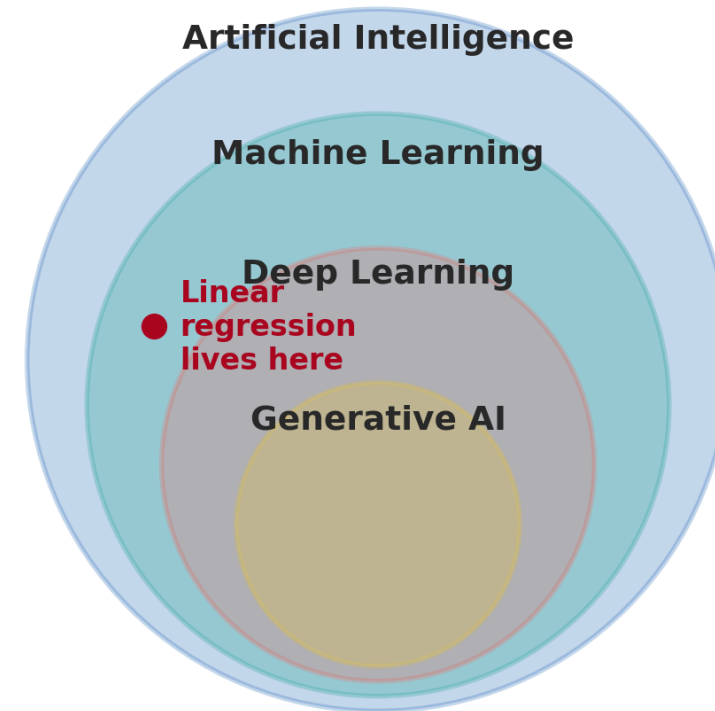
**Machine Learning** — a subset of AI.

Algorithms that detect patterns in large data sets and adapt, using supervised or unsupervised methods.

**Deep Learning** — a subset of ML that stacks layers of neural networks to extract high-level features from raw input.

**Generative AI** — a subset of DL that produces new content — text, images, code — from a prompt.

Each ring is a subset of the one outside it



# Regression as Supervised Learning

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Linear regression is the simplest **supervised learning** algorithm.

| Statistics term                                  | Machine learning term          |
|--------------------------------------------------|--------------------------------|
| Response variable $y$                            | Label / target                 |
| Predictor variable $x$                           | Feature                        |
| Regression slope $\beta_1$ / Intercept $\beta_0$ | Weight/coefficient / Bias term |
| Least squares                                    | Loss function minimization     |
| $R^2$                                            | Model performance metric       |
| Residual plot                                    | Error analysis / diagnostics   |

Every supervised ML model — from logistic regression to neural networks — generalizes the same idea: **find parameters that minimize prediction error on labeled training data.**

# Three Paradigms of Machine Learning

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## Supervised Learning

Learn from **labeled examples**

( $x \rightarrow y$  pairs)

- Regression — predict a number
- Classification — predict a category

## Unsupervised Learning

Find **structure without labels**

- Clustering — group similar items
- Dimensionality reduction (PCA)

## Reinforcement Learning

Learn by **trial and reward**

- Agent takes actions, receives feedback
- Game playing, robotics

# Deep Learning — What Makes It Different

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Traditional ML (including OLS regression):

- **You engineer the features** — you decide outdoor temperature is the predictor
- Model finds the best coefficients for those hand-crafted features

Deep Learning:

- **The network learns its own features** from raw data (pixels, tokens, signals)
- Stacked layers of transformations, each learning progressively abstract patterns



## Deep Learning — What Makes It Different (cont.)

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|                  | Traditional ML                         | Deep Learning                   |
|------------------|----------------------------------------|---------------------------------|
| Features         | Hand-crafted by domain expert          | Learned automatically from data |
| Data requirement | Hundreds to thousands of rows          | Millions to billions            |
| Interpretability | High ( $\beta_1$ has physical units)   | Low — "black box"               |
| CE use cases     | Regression, prediction, classification | Computer vision, NLP, control   |

# Large Language Models — Regression at Massive Scale

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An LLM is trained to **predict the next word (token)** — a supervised prediction problem.

The mathematical framework is identical to OLS:

- OLS: find  $\beta_0, \beta_1$  to minimize prediction error on  $(x, y)$  pairs
- LLM: find billions of weights to minimize prediction error on (text  $\rightarrow$  next word) pairs

|               | OLS Regression                        | Large Language Model                     |
|---------------|---------------------------------------|------------------------------------------|
| Parameters    | 2 ( $\beta_0, \beta_1$ )              | ~100 billion                             |
| Training data | 8,760 observations                    | Trillions of tokens                      |
| Training time | Milliseconds                          | Weeks on thousands of GPUs               |
| Output        | $\hat{y} = \beta_0 + \beta_1 \cdot x$ | Probability distribution over next token |

## Week 9 Quick Reference

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| Concept             | Formula                         |
|---------------------|---------------------------------|
| Regression equation | $\hat{y} = \beta_0 + \beta_1 x$ |
| Residual            | $e_i = y_i - \hat{y}_i$         |
| $R^2$ (from $r$ )   | $r^2$                           |
| Slope units         | units(y) / units(x)             |
| Predict at $x^*$    | $\beta_1 \times x^* + \beta_0$  |

### Tucson model numbers:

- $\beta_1 = 1.1585 \text{ kWh/}^\circ\text{F}$
- $\beta_0 = -60.68 \text{ kWh}$
- $R^2 = 0.601$
- Prediction at  $108^\circ\text{F} = 64.4 \text{ kWh/hr}$
- Prediction at  $75^\circ\text{F} = 26.2 \text{ kWh/hr}$

## Quick Check — Before the Lab

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45 seconds for A, 60 seconds for B

**Question A:** The regression model predicts cooling =  $-60.68 + 1.1585 \times 50 = -2.8$  kWh/hr at 50°F. Is this a problem? Why or why not?

**Question B:** If  $R^2 = 0.601$ , what percentage of cooling variance is NOT explained by outdoor temperature? Name one physical factor that likely accounts for part of it.

## Quick Check — Answers

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Before the lab

**Question A:**  $-2.8$  kWh/hr at  $50^{\circ}\text{F}$  is a **model limitation, not a calculation error** — cooling energy cannot go below zero (the chiller turns off). The model is valid above  $\sim 52^{\circ}\text{F}$ ; below that it extrapolates into impossible territory.

**Question B:** If  $R^2 = 0.601$ , then **39.9% of cooling variance is unexplained** by outdoor temperature alone. Physical factors that account for the residual:

- Occupancy schedule (lights, equipment, people)
- Solar gain and internal plug loads (cloud cover, servers, refrigeration)
- Building thermal mass (lag between outdoor  $T$  and indoor response)

  $R^2 = 0.601$  is solid for a single predictor; Week 10 adds occupancy and reaches  $R^2 = 0.910$ .

# Looking Ahead — Week 10: Multiple Regression

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**Today:** one predictor  $\rightarrow \hat{y} = \beta_0 + \beta_1 \cdot \text{Outdoor\_Temp\_F}$

**Week 10:** two predictors  $\rightarrow \hat{y} = \beta_0 + \beta_1 \cdot \text{Outdoor\_Temp\_F} + \beta_2 \cdot \text{BusinessHours}$

**New questions Week 10 answers:**

- After accounting for temperature, does occupancy explain additional variance?
- How much does the cooling baseline shift between occupied and unoccupied hours?
- Can adding more predictors hurt a model? (Adjusted  $R^2$  vs.  $R^2$ )

**Before next class:** in your lab notebook, predict whether adding `BusinessHours` will increase  $R^2$ . Write one sentence explaining your reasoning.